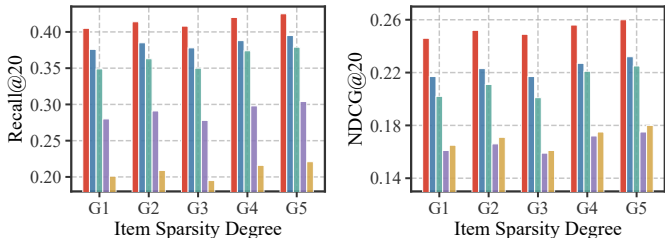


(a) Performance *w. r. t.* cold-start user groups



(b) Performance *w. r. t.* sparse item groups